B. Tech Degree VI Semester (Supplementary) Examination September 2010

ME 604 HEAT AND MASS TRANSFER

(2006 Scheme)

(Data book is permitted)

Time : 3 Hours

Maximum Marks : 100

PART A

(Answer all questions)

(8 x 5 =40)

I. (a) Hot milk at 65^0C is contained in a thin metal tumbler. In order to cool the milk two methods are proposed:
(i) Place the tumbler in a large vessel containing water at 25^0C
(ii) Stir the milk with a spoon.
Which method would you prefer for more rapid cooling of the milk and why ?

(b) Explain the importance of insulated tip solution for the fins used in practice.

(c) What is the Dittus - Boelter Equation? Where and when does it apply?

(d) Explain the criterion for deciding the type of convection (free or forced) in any given situation.

(e) Describe the phenomenon of radiation from real surfaces.

(f) Find the shape factor for a cylindrical cavity of diameter D and depth H with respect to itself.

(g) Explain the term 'number of heat transfer units'. Discuss various considerations essential for the design of heat exchangers.

(h) Write a note on the selection of the tube-side and shell-side fluid in heat exchangers.

PART B

(4 x 15 = 60)

II. (a) A steel tube carries steam at a temperature of 320^0C. A thermometer pocket of iron (K=52.3W/mk) of the diameter 1.5cm and 1mm thick is used to measure the temperature. The error to be tolerated is 1.5% maximum. Estimate the length of pocket necessary to measure the temperature within this error. The diameter of steel tube is 9.5cm. Suggest a suitable method of locating the thermometer pocket. The following data can be assumed. $h = 93W/m^2K$ Tube wall temperature = 120^0C. (8)

(b) A single cylinder 2 stroke IC engine is running at 800rpm. The maximum range of surface temperature oscillation recorded is 8^0C. Calculate the amount of heat absorbed by the cylinder wall during each power stroke. The bore and stroke of the piston are 80mm and 120mm respectively. Assume $K = 58.15\ W/m.k,\ \alpha = 0.05\ m^2/h.$ (7)

OR

III. (a) An electronic semi-conductor device has a rating of 60mW and for it's proper operation the inside temperature should not be more than 80^0C. It is found that the device can dissipate about 15mW of heat on it's own when placed in an environment at 40^0C. To avoid overheating of the device it is proposed to install aluminium (K= 190W/mK) square fins of size 0.5mm x 0.5mm and 1cm long to provide additional cooling. Find the number of fins required, taking $h=12.5W/m^2K$. (7)

(b) Heat is being transferred across a metal wall from a hot gas to cold liquid. The gas side co-efficient is $50W/m^2K$ and the liquid side co-efficient is $500W/m^2K$. It is desired to increase the heat transfer rate. However, it is not possible to alter the values of the heat transfer co-efficient on either side. It is therefore proposed to provide fins. On which side should the fins be provided and why? (4)

(c) Obtain an expression for the critical radius of insulation for a spherical shell. Give a physical explanation for the fact that a certain thickness of insulation may increase the rate of heat loss rather than reduce it. (4)

IV. (a) A metallic element (emissivity=1) of 6mm diameter is submerged horizontally in a water bath. Determine the power dissipation per unit length of the heater assuing it's surface temperature as 255^0C in stable film pool boiling. (7)

(b) Water flows through a tube of 22mm diameter with a velocity of 2m/s. Steam at 150^0C is being condensed on the outer surface of the tube thereby raising the temperature of water flowing inside the tube from 15^0C to 60^0C. Find the heat transfer co-efficient and the length of the tube required to meet the above requirement of heat. The resistance of tube and film may be neglected. (8)

OR

(P.T.O)

V. (a) A horizontal cylinder 25mm diameter and 60cm long is suspended in water at 20^0C. The cylinder surface is maintained at 55^0C. Estimate the rate of heat transfer between the cylinder and water. What would be the heat loss had this cylinder been suspended in light oil at 55^0C. The relevant properties of water and oil are:

Property	Water	Oil
P(Kg/m^3)	992	905
μ (Kg/h-m)	247	82
K(W/mK)	0.622	0.133
β (K^{-1})	3.96×10^{-4}	7.2×10^{-4}
Pr	4.64	324

Use $\overline{Nu_D} = 0.53(Ra_D)^{0.25}$. (9)

(b) Air at 1 atmosphere and 27^0C is forced through a 20mm diameter tube at an average velocity of 30cm/s. The tube wall is maintained at 127^0C. Estimate the heat transfer if the tube is 1m long. (6)

VI. A boiler furnace lagged with plate steel ($\in = 0.6$) is laid with fireclay bricks ($\in = 0.8$) on the inside. The distance between the lagging and setting brick is 300mm and it may be assumed small compared with the size of the furnace. Calculate the loss of heat per unit area between the lagging and setting if these are at 400K and 323K respectively.

A radiation shield of emissivity 0.6 is now introduced between the brick setting and lagging of the furnace. Compute the radient flux between these and the loss of heat by radiation to surroundings.

If it is desired to restrict the loss of heat by radiation to 60W/m^2, calculate the emissivity of the radiation shield to be introduced between the furnace setting and steel lagging to meet this requirement. (15)

OR

VII. A 10cm OD steel pipe carrying steam at 4.905×10^5 Pa is lagged with 5cm thick insulation. The surrounding air temperature is 27^0C. The conductivity and emissivity of the insulating material are 0.14W/mK and 0.92 respectively. If the heat transfer co-efficient on the outer surface of the lagged pipe is 11.63W/m^2 ^{0}C. Calculate, neglecting the pipe resistance and internal boundary resistance.
(i) heat loss per metre length of the pipe by convection
(ii) heat loss per metre length of the pipe by radiation
(iii) radiation and overall heat transfer co-efficients
(iv) rate of condensation for 50m pipe length. (15)

VIII. Water at the rate of 4 Kg/S is heated from 38^0C to 55^0C in a shell and tube type heat exchanger. The water is to flow inside tubes of 2cm diameter with an average velocity of 35cm/S. Hot water available at 95^0C and at the rate of 2Kg/S is used as the heating medium on the shell side. If the length of the tubes must not be more than 2m calculate the number of tube passes, the number of tubes per pass and the length of the tubes for one pass shell, assuming U_0=1500W/m^{2}K. (15)

OR

IX. Two identical single pass counter-flow heat exchangers are used for heating water at 30^0C with the help of hot oil (C_p=2.1KJ/Kg.K) at 100^0C. The mass flow rates of the water and oil are 50 and 100 Kg/min respectively. The heat exchangers are connected in series on the water side and in parallel on the oil side, the oil flow being split equally between the two heat exchangers at the inlet and joined up later at the exit as shown in figure. Assuming that for each heat exchanger U=350W/m^{2}K and A=10m^2, calculate the exit temperatures of the water and the oil. (15)